

Rapport de stage de fin d'études

Rapport de Stage Ing5 - 3^{ème} année Cycle Ingénieur Promotion 2026

Nom :

Durand

Prénom :

Raphaël

Majeure:

- ☐ DATA & IA
- ☐ CREATE
- ☐ DEF
- ☐ EE
- ☐ FIN
- ☒ SANTE
- ☐ SI/CYB/CLOUD
- ☐ SE/VCA
- ☐ SEN

Entreprise d'accueil

Nom : Chipiron

Adresse : 2|8 Rue Gaston Rebuffat, Paris 75019

Adresse du lieu de stage si différent :

Engagement de Confidentialité : ☐ oui ☒ non

Reçu le 09 / 02 / 2026

Rapport Confidentiel à remettre au tuteur de stage à l'issue de la soutenance : ☐ oui ☒ non

Pénalités Observées :

Description de la mission :

This internship will focus on developing and validating both (1) acceleration/undersampling strategies paired with (2) deep learning reconstruction to recover high-quality contrast-enhanced MRI from a non-contrast image and a strongly accelerated, low SNR contrast-enhanced acquisition.

- Literature Review: Review contemporary literature in contrast synthesis, image-to-image translation, multi-contrast reconstruction, and accelerated MRI, with emphasis on methods combining multi-contrast inputs.
- Development and Integration: Design and implement deep learning models to recover diagnostic-grade contrast-enhanced images. Explore strategies to jointly optimize k-space acceleration and reconstruction quality.
- Data Processing and Evaluation: Develop pipelines for data harmonization, pre-processing, quantitative evaluation, and validation against reference fully sampled contrast-enhanced datasets.